

Ravin Patra

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Professional Summary

- Experienced Software Engineer with 7 years of experience across Finance, Manufacturing, and Defense domains.
- Specialized in building cloud-native platforms using microservices, serverless architectures, and event-driven systems on AWS with services such as Lambda, S3, SQS, SNS, Step Functions, Glue, and DynamoDB.
- Strong experience in Treasury and payment systems, including ACH, Wire, and Check processing, payment lifecycle management, and financial data integrations.
- Proficient in Python, Java, ETL pipelines, RESTful API development, AWS CDK, and CI/CD automation using GitHub Actions.
- Experienced in modernizing legacy enterprise systems by migrating them to scalable cloud-native architectures.
- Engineered domain-specific AI agents within the VS Code ecosystem to streamline development lifecycle.
- Provide technical leadership in platform development, collaborating with cross-functional teams and vendor engineers to deliver scalable financial systems.
- Passionate about designing scalable, resilient, and high-performance backend systems that improve operational efficiency and system reliability.

Experience

Principal Financial Group , Sr. Software Engineer

Jul 2025 – Present

- Leading the design and development of a Financial Foundations platform for enterprise ledger transformation, migrating financial data from PeopleSoft Ledger to Oracle Ledger.
- Architected a serverless AWS data processing platform using S3, Lambda, SQS, SNS, Step Functions, and Glue to ingest, validate, and transform financial data across multiple business units.
- Designed scalable, event-driven microservices to support high-volume financial data processing while ensuring reliability, auditability, and fault tolerance.
- Serve as technical lead, coordinating with vendor development teams, conducting implementation reviews, and enforcing alignment with architectural and coding standards.
- Collaborate with cross-functional teams including finance, treasury, and infrastructure to translate business requirements into secure and scalable technical solutions.
- Conduct code reviews, resolve defects, and mentor developers while maintaining 80%+ unit and integration test coverage.
- Developed PoCs leveraging GenAI, NLP, and RAG to improve observability and compliance; engineered an AI-powered agent that autonomously generates production-grade AWS architecture diagrams on every commit, ensuring architectural documentation remains continuously up-to-date without manual intervention.
- Designed and maintained custom AI agent workflows integrated into VS Code to automate repetitive engineering tasks, improving coding productivity and consistency across the team.

Principal Financial Group , Software Engineer

Dec 2021 – Jul 2025

- Designed end-to-end integration to transmit payment data to Kyriba, supporting Wire, ACH, and Check payment processing.
- Implemented ISO 20022 compliant payment file structures with automated reconciliation and duplicate detection to strengthen financial data integrity.
- Led React-based UI modernization integrating Financial Institution (FI) APIs to retrieve and display check images, enhancing the end-to-end payment processing workflow.
- Automated manual payment workflows, significantly improving operational efficiency and reducing manual intervention.
- Strengthened system security and observability by implementing OAuth 2.0 authentication (Apigee, Okta, Azure

- Entra ID) and centralized monitoring dashboards using CloudWatch, Grafana, ELK, and Dynatrace.
- Managed infrastructure as code using AWS CDK and streamlined CI/CD pipelines using GitHub Actions.

Tata Consultancy Services, System Engineer

July 2019 – Dec 2021

- Collaborated with various GE clients (Aviation, Digital, and MMF Plant)
- Migrated on-premises applications to Windows Server and AWS Cloud using EC2, lambda and Fargate for compute, Route 53 for DNS, and VPC for secure network architecture
- Utilized various cloud technologies (Cloud Foundry, Predix, AWS, GCP) to modernize legacy applications.
- Developed new dashboard interfaces using React, PolymerJs and VueJs based on client requirements, including Work In Progress dashboards, BOM trackers, Material QR generators, and INCCID.
- Designed application to manage invoices and keep track of BRC.

Personal Projects

Multi-Repo Architecture Intelligence Agent

In Progress

- Developing an AI-powered agent that ingests multiple GitHub repositories and generates a unified architecture diagram, mapping inter-repo dependencies, service connections, and system boundaries with contextual detail.
- Designed to scale across enterprise codebases where multiple services interact, enabling engineers to visualize the full system landscape without manual documentation effort.

Self-Improving Natural Language to DB2 Query Engine

In Progress

- Building a LLM-free conversational query engine for DB2 that uses metadata, embeddings, and vector search to identify relevant tables, parameters, and relationships from natural language input — eliminating dependency on external AI APIs.
- Implemented a user feedback loop that continuously refines the underlying metadata after every query interaction, improving table and parameter resolution accuracy over time without retraining.
- Actively integrating ML concepts from ongoing MTech research to enhance query inference, entity resolution, and recommendation performance.

Skills

Languages: Python, Java, SQL, T-SQL, NoSQL, JavaScript/TypeScript.

Cloud & Infrastructure: AWS (Lambda, S3, SQS, SNS, Step Functions, Glue, DynamoDB, CDK), Cloud Foundry.

Frameworks & Libraries: Spring Boot, Spring MVC, React.

AI & Agentic Systems: Agentic AI for SDLC, MCP (Model Context Protocol) Servers, VS Code AI Agents, GitHub Copilot, GenAI (RAG, NLP).

Monitoring: ELK Stack, Dynatrace, Grafana, CloudWatch.

Tools & Utilities: Agile (Jira), Vault (HashiCorp, BeyondTrust), Version Control (Git, BitBucket), CI/CD (GitHub Actions), Code Security (CodeQL, Fortify, SonarQube), FTP Tools (Cleo, GoAnywhere), SSMS.

Education

Master of Technology, Artificial Intelligence & Machine Learning

April 2026 – March 2028

Birla Institute of Technology and Science (BITS), Pilani

Bachelor's of Engineering, Computer Engineering

June 2015 – June 2019

Padmabhooshan Vasantdada Patil Institute of Technology, Pune (CGPA: 8.59/10.00)

Personal Information

- Languages Known:** English, Hindi, Marathi, Oriya
- Permanent Address:** Flat no 81, bldg no 3, Soba Puram, Warje, Pune - 411058